

Hi,

This website, UK Travel Stories, was created as a demonstration project to perform a comprehensive Core Web Vitals audit and implementation process.

The site intentionally includes several technical performance issues such as slow-loading elements, layout instability, and optimization gaps. These issues were deliberately left unresolved so they could be identified, analyzed, documented, and then addressed through a structured audit, recommendation, and implementation workflow.

The purpose of this project is to demonstrate my ability to:

- Conduct a full Core Web Vitals audit using industry-standard tools
- Identify critical performance issues affecting user experience and SEO
- Provide practical, data-driven recommendations and solutions
- Apply technical SEO and performance optimisation techniques
- Implement fixes based on audit findings
- Deliver a before-and-after performance improvement report

This project simulates a real-world Core Web Vitals optimisation scenario, where a website presents multiple performance improvement opportunities. The findings, recommendations, and implementation results demonstrate how such issues can be diagnosed, resolved, and improved to enhance both website speed and user experience.

Thank you.

PART 1

Core Web Vitals Audit Report

UK Travel Stories · May 2026

Core Web Vitals Audit Report

Performance Analysis & Optimisation Strategy

CLIENT

UK Travel Stories

PREPARED BY

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DATE

May 2026

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Executive Summary

Mobile Scores (Before Optimisation)

70 Performance	90 Accessibility	96 Best Practices	82 SEO
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Desktop Scores (Before Optimisation)

95 Performance	91 Accessibility	96 Best Practices	82 SEO
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The May 2026 Lighthouse audit of UK Travel Stories identified significant performance issues, particularly on mobile. The mobile Performance score was 70 and desktop was 95 — both driven by the same underlying technical deficiencies, with mobile experiencing greater impact due to network and CPU throttling.

The audit identified 13 distinct issues across four categories — Performance, Accessibility, SEO, and Best Practices. The most critical finding was that hero content animations were configured to start at opacity:0, rendering all above-the-fold content invisible to the browser's LCP measurement for over 1.2 seconds. Combined with render-blocking font requests, an unpreloaded hero image, and broken CSS rules, these issues created a compounding effect that pushed mobile LCP to 7.2 seconds — nearly three times above the 2.5-second good threshold.

In addition to performance issues, the audit uncovered accessibility contrast failures, a skipped heading hierarchy, missing SEO metadata, a non-crawable navigation link, and three missing image tags that left key content sections visually empty.

Objectives

The audit was conducted to establish a measurable baseline and identify the technical root causes behind the underperforming load experience for UK Travel Stories. The specific objectives were:

- Measure all Core Web Vitals signals including LCP, FCP, CLS, TBT, and Speed Index.
- Identify every performance bottleneck blocking the critical render path.
- Uncover accessibility issues that reduce usability and fail WCAG standards.
- Identify SEO deficiencies that reduce search visibility and crawlability.
- Document all findings with specific code-level evidence and actionable recommendations.
- Establish a clear priority order for remediation based on impact.

The audit was designed to simulate a real client engagement, presenting findings in a way that can be directly acted upon within a static GitHub Pages environment without requiring backend infrastructure or build tooling.

Website Overview

UK Travel Stories is a static travel storytelling website hosted on GitHub Pages at <https://athiramolps.github.io/uktravelstories/>. The site features a full-viewport hero section with a background image, a magazine-style card grid of five destination articles, an about section, a contact form, and a footer.

The site is built entirely with HTML, CSS, and vanilla JavaScript — no frameworks, no build pipeline. Every performance gain must therefore come from front-end optimisation at the asset and markup level. The page relied heavily on external Unsplash images and Google Fonts, both of which were identified as major contributors to the slow load experience.

Baseline Performance

The following metrics were recorded using Lighthouse on a simulated Moto G Power device with Slow 4G throttling (mobile) and an unthrottled desktop environment. Both sets of results are provided as the baseline before any optimisation was applied.

Mobile Baseline

Metric	Value	Threshold (Good)	Status
Performance Score	70	90+	Needs Improvement
First Contentful Paint	3.2s	≤ 1.8s	Poor
Largest Contentful Paint	7.2s	≤ 2.5s	Poor
Cumulative Layout Shift	0.014	≤ 0.1	Good
Speed Index	3.3s	≤ 3.4s	Near Threshold
Total Blocking Time	0 ms	≤ 200ms	Good
SEO Score	82	90+	Needs Improvement
Accessibility Score	90	90+	Good

Desktop Baseline

Metric	Value	Threshold (Good)	Status
Performance Score	95	90+	Good
First Contentful Paint	0.8s	≤ 1.8s	Good
Largest Contentful Paint	1.4s	≤ 2.5s	Good
Cumulative Layout Shift	0.007	≤ 0.1	Good
Speed Index	0.9s	≤ 3.4s	Good
Total Blocking Time	0 ms	≤ 200ms	Good
SEO Score	82	90+	Needs Improvement
Accessibility Score	91	90+	Good

Issues Identified

The audit identified 13 issues across four categories. Each issue is documented below with its root cause, impact, and recommended fix.

Performance

Issue 1 — Render-Blocking Google Fonts

What was wrong:

The Google Fonts stylesheet was loaded with `rel="stylesheet"`, placing it on the critical render path. The browser could not paint anything until all three font families (13 weight variants, approximately 140 KiB) were fully downloaded. Lighthouse estimated a saving of approximately 750ms.

How it was fixed:

Defer font loading using `media="print"` combined with `onload="this.media='all'"`. This removes the stylesheet from the critical path, allowing the page to paint immediately using system fallback fonts while custom fonts load in the background. Font weights were also reduced from 13 variants to 7.

```
BEFORE: <link href="fonts.googleapis.com/css2?...wght@0,400;0,600;0,700;0,900" rel="stylesheet"/>
```

```
AFTER: <link href="...wght@0,700;0,900;1,400" media="print" onload="this.media='all'"/>
```

Issue 2 — Hero Image Not Preloaded

What was wrong:

The hero background image was applied via CSS `background-image`. The browser's preload scanner cannot detect images referenced in stylesheets, so the image was only discovered after the CSS was fully parsed — very late in the loading sequence.

How it was fixed:

Add a preload link tag with `fetchpriority=high` inside head so the browser begins downloading the hero image as the very first network request, before any CSS or JS is processed.

```
ADDED: <link rel="preload" as="image" href="/tower.webp" fetchpriority="high"/>
```

Issue 3 — Premature Style Block Before Meta Charset

What was wrong:

A style block appeared in head before the meta charset=UTF-8 declaration. The browser requires the charset declaration as the first statement to correctly decode the file. Any content before it forces the parser to restart.

How it was fixed:

Delete the stray style block and ensure meta charset=UTF-8 is the very first tag inside head.

```
REMOVED: <style>.hero-bg{ background-image:url('/tower.webp'); }</style>
(appared BEFORE <meta charset> – caused parser restart)
```

Issue 4 — Hero Background CSS Missing All Properties

What was wrong:

The `.hero-bg` CSS rule only contained `background-image`. It was missing `position:absolute`, `inset:0`, `background-size:cover`, `background-position:center`, and the zoom animation. The hero image was not covering the

viewport.

How it was fixed:

Restore the complete .hero-bg rule with all required layout and visual properties.

```
BEFORE: .hero-bg { background-image: url('./tower.webp'); }

AFTER: .hero-bg { position:absolute; inset:0; background-image:url('./tower.webp');
background-size:cover; background-position:center;
will-change:transform;
animation:hzoom 20s ease-in-out infinite alternate; }
```

Issue 5 — Hero Animations Starting at opacity:0 — Main LCP Killer

What was wrong:

The @keyframes fup animation was defined as from{opacity:0; transform:translateY(28px)} with fill-mode:both. This made every hero element completely invisible for up to 1,180ms after page load. Because the browser cannot record an LCP paint for an invisible element, LCP was artificially pushed to 7.2 seconds.

How it was fixed:

Change the animation's from state to start at opacity:1, removing the opacity fade. The elements remain visible immediately and LCP is measured as soon as the content is rendered.

```
BEFORE: @keyframes fup { from{opacity:0; transform:translateY(28px);} to{opacity:1;
transform:translateY(0);} }

AFTER: @keyframes fup { from{opacity:1; transform:translateY(28px);} to{opacity:1;
transform:translateY(0);} }
```

Issue 12 — Forced Reflow in JavaScript

What was wrong:

Inside the openArticle() function, ov.classList.add('open') was called first (a DOM write), immediately followed by reading ov.scrollTop (a DOM read). After any DOM write invalidates the layout tree, any subsequent DOM read forces the browser to immediately recalculate the full page layout — an expensive synchronous forced reflow.

How it was fixed:

Swap the order: read scrollTop first (while layout is still valid), then write with classList.add. This eliminates the forced reflow entirely.

```
BEFORE: ov.classList.add('open'); // write – invalidates layout
ov.scrollTop = 0; // read – forces full reflow

AFTER: ov.scrollTop = 0; // read – layout still valid
ov.classList.add('open'); // write – no reflow triggered
```

Issue 13 — Missing Story Card Image Tags

What was wrong:

Three story cards (Tower of London, Edinburgh, Bath) were missing their img tags entirely. The Tower of London card is the primary above-the-fold content and the intended LCP element — without its image tag, LCP was measured incorrectly.

How it was fixed:

Restore all three img tags. The Tower of London card received fetchpriority=high to signal it as the LCP element. Edinburgh and Bath cards received loading=lazy as they are below the fold.

BEFORE: <!-- No img tag for Tower of London card -->

AFTER:

SEO

Issue 6 — Missing Meta Description

What was wrong:

No meta name=description tag was present in the document. Google uses the meta description in search result snippets. Its absence directly lowered the SEO score from 100 to 82.

How it was fixed:

Add a unique, descriptive meta description inside head that accurately summarises the page content.

ADDED: <meta name="description" content="UK Travel Stories - an independent UK travel journal."/>

Issue 7 — Uncrawlable Home Navigation Link

What was wrong:

The Home link in the navbar had no href attribute — only an onclick JavaScript handler. Search engine crawlers cannot follow onclick-only links, meaning the homepage could not be discovered through the navigation.

How it was fixed:

Add href=# to the anchor element so it is a valid, crawlable link.

BEFORE: Home

AFTER: Home

Accessibility

Issue 8 — Low Contrast: Tag Pill Text

What was wrong:

The tag pill element used terra colour #bf5b38 as text on a parchment background #f0ebe0. The contrast ratio was only 3.39:1. WCAG AA requires a minimum of 4.5:1 for small text.

How it was fixed:

Change the tag pill text colour from --terra to --navy (#0d1f35), which achieves a contrast ratio of 13.5:1 — well above the WCAG AA requirement.

BEFORE: .tag-pill { color: var(--terra); } /* #bf5b38 — contrast 3.39:1 — FAILS */

AFTER: .tag-pill { color: var(--navy); } /* #0d1f35 — contrast 13.5:1 — PASSES */

Issue 9 — Low Contrast: Primary Button

What was wrong:

The Read Our Stories button used white text on a terra background (#bf5b38). The contrast ratio was 3.99:1. At the button's font size of 0.93rem, WCAG AA requires 4.5:1. The hover colour #a84e30 was also lighter than the base, making hover visually incorrect.

How it was fixed:

Introduce a darker terra variable --terra-dk: #9e3d1e which achieves a contrast ratio of 5.9:1 with white text. Set the hover colour to #7d2e0f (darker).

```
BEFORE: .btn-p { background: var(--terra); } /* 3.99:1 – FAILS */
BEFORE: .btn-p:hover { background: #a84e30; } /* lighter than base – wrong */

ADDED: --terra-dk: #9e3d1e;
AFTER: .btn-p { background: var(--terra-dk); } /* 5.90:1 – PASSES */
AFTER: .btn-p:hover { background: #7d2e0f; } /* darker than base – correct */
```

Issue 10 — Heading Hierarchy Skipping H3

What was wrong:

The About section used h2 for its main heading then jumped directly to h4 for the four feature cards, completely skipping the h3 level. Screen readers and accessibility tools rely on sequential heading structure.

How it was fixed:

Change all four feature card headings from h4 to h3 and update the corresponding CSS selector from .fc h4 to .fc h3.

```
BEFORE: <h4>Curated Destinations</h4> CSS: .fc h4 { ... }

AFTER: <h3>Curated Destinations</h3> CSS: .fc h3 { ... }
(Applied to all 4 feature headings)
```

Best Practices

Issue 11 — Bath Card Image Returning 404

What was wrong:

The Bath destination card's img tag pointed to a deleted Unsplash photo (photo-1569336415962-a4bd9f69ce07). The image returned a 404 Not Found error and caused the Bath card to display with no thumbnail image.

How it was fixed:

Replace the broken URL with a verified working Unsplash image of Bath.

```
BEFORE: photo-1569336415962-a4bd9f69ce07 (404 – image deleted from Unsplash)

AFTER: photo-1521106581851-da5b6457f674 (200 – verified working image)
```

Conclusion

The audit established a clear picture of why UK Travel Stories was underperforming on mobile. The site had a strong visual concept and stable layout, but the technical delivery contained multiple compounding issues that prevented it from being measured accurately or loading efficiently.

The most significant finding was that the hero animations were configured in a way that made all above-the-fold content invisible during the measurement window. This single issue alone was responsible for the majority of the LCP penalty. Combined with render-blocking fonts, a missing image preload, and broken CSS rules, the conditions for a poor Core Web Vitals result were thoroughly established.

The 13 issues identified represent a complete optimisation roadmap. Once implemented, the site is expected to achieve substantial improvements across all four Lighthouse categories. The implementation findings are documented in Part 2 of this report.

Glossary

Term	Definition
First Contentful Paint (FCP)	The time at which the browser first renders text or an image, giving the user the first visible sign that the page is loading.
Largest Contentful Paint (LCP)	The render time of the largest visible image or text block in the viewport. The primary loading metric in Core Web Vitals. Good threshold: $\leq 2.5s$.
Cumulative Layout Shift (CLS)	Measures unexpected movement of visible page elements during load. Lower is better. Good threshold: ≤ 0.1 .
Total Blocking Time (TBT)	Sum of all periods between FCP and Time to Interactive when the main thread was blocked for more than 50ms.
Speed Index	A Lighthouse metric showing how quickly the visible contents of a page are populated during load.
Render-Blocking Resource	A CSS or JavaScript file that prevents the browser from painting content until it has been fully downloaded and processed.
Critical CSS	The minimum CSS needed to render above-the-fold content, allowing the first paint to occur without waiting for the full stylesheet.
Forced Reflow	A performance penalty caused by reading layout properties immediately after writing to the DOM, forcing the browser to recalculate layout synchronously.
fetchpriority	An HTML attribute that instructs the browser to treat a resource as high priority, starting its download earlier in the loading sequence.
WCAG AA	Web Content Accessibility Guidelines Level AA. The standard requiring a minimum colour contrast ratio of 4.5:1 for normal text.

THANK YOU

Core Web Vitals Audit Report

CLIENT

UK Travel Stories

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PART 2

Core Web Vitals Implementation Report

UK Travel Stories · May 2026

Core Web Vitals Implementation Report

Performance Optimisation & Results

CLIENT

UK Travel Stories

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DATE

May 2026

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Executive Summary

Following the audit documented in Part 1, all 13 identified issues were implemented. The results demonstrate a dramatic and measurable transformation in performance. The mobile Performance score improved from 70 to 100, LCP dropped from 7.2 seconds to 1.7 seconds, and FCP improved from 3.2 seconds to 1.0 second. The SEO score reached 100, and Best Practices achieved a perfect 100.

Post-Optimisation Scores — Mobile

100 Performance	93 Accessibility	100 Best Practices	100 SEO
---------------------------------------	--	--	-------------------------------

Post-Optimisation Scores — Desktop

98 Performance	94 Accessibility	100 Best Practices	100 SEO
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The implementation phase applied front-end best practices within the constraints of a static GitHub Pages environment — no server-side changes, no build pipeline, no framework migration. Every fix was applied directly to the HTML, CSS, and JavaScript source.

Objectives

The implementation objectives were directly aligned to the 13 issues identified in the audit:

- Eliminate every render-blocking resource from the critical path.
- Ensure the LCP element is visible and measurable from the moment the page loads.
- Preload the hero image so the browser begins fetching it as early as possible.
- Restore all missing image tags and correct all broken asset URLs.
- Fix all WCAG AA colour contrast failures to improve accessibility.
- Correct the heading hierarchy so screen readers can navigate the page structure.
- Add missing SEO metadata and make all navigation links crawlable.
- Remove JavaScript forced reflow to improve main thread efficiency.
- Keep all solutions practical and maintainable within a static site environment.

Tools Used

Tool	Purpose	How Used
Google PageSpeed Insights	Performance scoring and metric measurement	Establish baseline and record post-optimisation scores on both mobile and desktop.
Lighthouse (Chrome)	Audit diagnostics and recommendations	Identify render-blocking resources, image issues, JS problems, SEO and accessibility gaps.
Chrome DevTools	Waterfall, coverage, and console inspection	Pinpoint exact sources of delay, find broken requests, inspect animation timing.
cwebp / image pipeline	Asset conversion and compression	Convert JPEG images to WebP format at correct display dimensions.
HTML / CSS / JS standards	Static optimisation techniques	Critical CSS analysis, animation fixes, semantic link rewrites, heading restructure.

Website Overview

UK Travel Stories is a content-led travel website with a high visual emphasis. It uses a hero section, destination features, story cards, and footer/contact sections to create a magazine-style presentation. Because the site is image-heavy, its user experience is strongly influenced by how quickly the browser can locate, download, decode, and paint the most visible assets.

The site is hosted as a static project on GitHub Pages at <https://athiramolps.github.io/uktravelstories/>, which means all optimisations must be applied at the front-end level without relying on server-side processing. The implementation strategy therefore prioritises the highest-impact above-the-fold resources first.

Baseline Performance

Before optimisation, the site showed a mobile Performance score of 70 and desktop score of 95. Both were affected by the same underlying issues, with mobile experiencing greater impact due to device CPU and network throttling. The diagnostic output highlighted approximately 2,120 ms of estimated savings from render-blocking requests and more than 500 KiB of potential savings from image delivery alone.

Mobile Baseline

Metric	Baseline Value	Target (Good)	Status
Performance Score	70	90+	Needs Improvement
First Contentful Paint	3.2s	≤ 1.8s	Poor
Largest Contentful Paint	7.2s	≤ 2.5s	Poor
Cumulative Layout Shift	0.014	≤ 0.1	Good
Speed Index	3.3s	≤ 3.4s	Near Threshold
Total Blocking Time	0 ms	≤ 200ms	Good
SEO Score	82	90+	Needs Improvement
Accessibility Score	90	90+	Good

Desktop Baseline

Metric	Baseline Value	Target (Good)	Status
Performance Score	95	90+	Good
First Contentful Paint	0.8s	≤ 1.8s	Good
Largest Contentful Paint	1.4s	≤ 2.5s	Good
Cumulative Layout Shift	0.007	≤ 0.1	Good
Speed Index	0.9s	≤ 3.4s	Good
Total Blocking Time	0 ms	≤ 200ms	Good
SEO Score	82	90+	Needs Improvement
Accessibility Score	91	90+	Good

Implementation Phase

The 13 fixes were applied in priority order, starting with the issues that had the most direct impact on LCP and FCP. Each step is described below, grouped by category.

Performance

Step 1 — Fix Hero Animation (Issue 5)

This was applied first as it was the single biggest cause of the LCP failure. The `@keyframes fup` animation was changed from `from{opacity:0}` to `from{opacity:1}`. This one change alone is estimated to have reduced LCP by approximately 1.2 seconds, as the browser could now record the LCP paint the moment the hero content was rendered.

Step 2 — Restore Missing Image Tags (Issue 13)

The featured Tower of London card was given back its `img` tag with `fetchpriority=high`, ensuring the browser treats it as the highest-priority image request. The Edinburgh and Bath cards received their images with `loading=lazy` as they are below the primary view.

Step 3 — Fix the hero-bg CSS Rule (Issue 4)

The `.hero-bg` CSS rule was restored with all required properties: `position:absolute`, `inset:0`, `background-size:cover`, `background-position:center`, and the `hzoom` animation. Without these the hero image did not fill the viewport.

Step 4 — Defer Google Fonts and Reduce Weights (Issue 1)

The Google Fonts link was changed to use `media=print` with `onload`, removing it from the critical render path. Font variants were reduced from 13 to 7, cutting the font payload from approximately 140 KiB to approximately 80 KiB. A `noscript` fallback was added for browsers with JavaScript disabled.

Step 5 — Add Hero Image Preload (Issue 2)

A preload link tag with `fetchpriority=high` was added to head pointing to the `tower.webp` file. This ensures the browser begins downloading the hero image on the very first network request before CSS is even parsed.

Step 6 — Fix Head Structure (Issue 3)

The stray style block that appeared before `meta charset=UTF-8` was removed. The meta charset tag was confirmed as the first element in head, preventing the parser restart that was adding unnecessary overhead to FCP.

Step 12 — Fix JavaScript Forced Reflow (Issue 12)

Inside `openArticle()`, the order of DOM read and write was swapped. `ov.scrollTop = 0` is now called before `ov.classList.add('open')`, ensuring the layout read happens while the layout tree is still valid and no forced synchronous reflow is triggered.

SEO

Step 7 — Add Meta Description (Issue 6)

A descriptive meta description was added to head. This is required for the SEO score to reach 100 and provides Google with content for search result snippets.

Step 8 — Fix Uncrawlable Home Link (Issue 7)

href=# was added to the Home navigation anchor, making it a valid crawlable link. return false was added to the onclick handler to prevent the default scroll from conflicting with the custom smooth scroll logic.

Accessibility

Step 9 — Fix Colour Contrast Issues (Issues 8 and 9)

The tag pill text colour was changed from --terra (#bf5b38, contrast 3.39:1) to --navy (#0d1f35, contrast 13.5:1). A new --terra-dk variable (#9e3d1e, contrast 5.9:1) was introduced for buttons and applied to .btn-p, .nav-cta, and their hover states. The hover colour was corrected to be darker than the base.

Step 10 — Fix Heading Hierarchy (Issue 10)

All four h4 feature card headings in the About section were changed to h3. The CSS selector .fc h4 was updated to .fc h3. The page now has a correct sequential heading structure: H1 (page title) > H2 (section titles) > H3 (feature cards).

Best Practices

Step 11 — Fix Bath Image 404 (Issue 11)

The broken Unsplash URL for the Bath card image was replaced with a verified working alternative. The same working URL was applied to both the card thumbnail and the hero image in the JavaScript DATA array to ensure the full article view also loads correctly.

Post-Optimisation Performance

Mobile Results

Metric	Before	After	Change	Result
Performance Score	70	100	+30	Huge improvement
Accessibility	90	93	+3	Improved
Best Practices	96	100	+4	Perfect
SEO	82	100	+18	Major improvement
LCP	7.2s	1.7s	-5.5s	Massive improvement
FCP	3.2s	1.0s	-2.2s	Much faster
Speed Index	3.4s	1.0s	-2.4s	Very fast
CLS	0.014	0.015	≈0	Still excellent
TBT	0ms	0ms	0	Perfect

Desktop Results

Metric	Before	After	Change	Result
Performance Score	95	98	+3	Improved
Accessibility	91	94	+3	Improved
Best Practices	96	100	+4	Perfect
SEO	82	100	+18	Major improvement
LCP	1.4s	0.8s	-0.6s	Excellent
FCP	0.8s	0.8s	0	Same
Speed Index	0.9s	1.2s	+0.3s	Within range
CLS	0.007	0.013	≈0	Still excellent
TBT	0ms	0ms	0	Perfect

Impact Summary

The implementation produced exceptional results across every measured dimension.

Performance

The mobile Performance score went from 70 to a perfect 100 — a 30-point gain. LCP fell from 7.2 seconds (classified as Poor) to 1.7 seconds (Good), a reduction of 76%. FCP fell from 3.2 seconds to 1.0 second, a reduction of 69%. Speed Index fell from 3.4 seconds to 1.0 second.

The single highest-impact change was fixing the animation opacity issue (Issue 5). By allowing the LCP element to be visible from the moment of render rather than after a 900ms+ delay, the browser was able to record LCP accurately for the first time. Combined with the image preload and font deferral, the combined effect moved the site from Poor to the Good range for every loading metric.

SEO

The SEO score improved from 82 to a perfect 100 on both mobile and desktop. The two fixes responsible were adding the missing meta description and making the Home navigation link crawlable. Both were straightforward single-line changes with significant scoring impact.

Accessibility

Accessibility improved from 90 to 93 on mobile and from 91 to 94 on desktop. The contrast ratio for tag pill text improved from 3.39:1 (failing WCAG AA) to 13.5:1. The button contrast improved from 3.99:1 to 5.9:1. The heading hierarchy was corrected so screen readers can now navigate the page in sequential order.

Best Practices

Best Practices reached a perfect 100 on both mobile and desktop, up from 96. The primary fix was resolving the 404 console error from the broken Bath image URL, which was the only remaining flagged issue in this category.

Conclusion

The Core Web Vitals optimisation project for UK Travel Stories demonstrates that significant performance improvements can be achieved through targeted front-end changes, without requiring a change of hosting platform, build pipeline, or framework.

The audit process systematically uncovered 13 issues that were individually contributing to poor scores. The most impactful discovery — that hero animations were hiding content from LCP measurement — illustrates how a single CSS property can produce a disproportionate effect on measured performance. Once identified and corrected, along with the supporting fixes for fonts, image loading, and broken assets, the page achieved scores that represent best-in-class performance.

The project now serves as a complete and demonstrable case study: a site with intentional technical deficiencies was audited using industry-standard tooling, all issues were documented with code-level evidence and prioritised by impact, fixes were applied methodically in a static environment, and the results were verified with a post-implementation Lighthouse run achieving a perfect mobile Performance score of 100.

THANK YOU

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